

TMUA Formula + Fact Sheet

A compact but specification-aware reference sheet for TMUA Paper 1 and Paper 2. Use it for recall, but practise applying these under timed conditions.

**Exam reminder:** There is no formula booklet in TMUA. Paper 1 tests mathematical application. Paper 2 tests mathematical reasoning and proof, using Sections 1 knowledge plus logic and argument skills. There is no negative marking, so attempt every question.

1. Algebra, indices, surds and polynomials

**INDICES**  
Index laws:  $a^m a^n = a^{m+n}$ ,  $\frac{a^m}{a^n} = a^{m-n}$   
 $(a^m)^n = a^{mn}$ ,  $(ab)^n = a^n b^n$ ,  $a^{-n} = \frac{1}{a^n}$   
 $a^{1/n} = \sqrt[n]{a}$ ,  $a^{m/n} = \sqrt[n]{a^m}$   
*Think: use rational, negative and fractional indices frequently.*

**SURDS**  
 $\sqrt{ab} = \sqrt{a}\sqrt{b}$ ,  $\sqrt{\frac{a}{b}} = \frac{\sqrt{a}}{\sqrt{b}}$   
 $\sqrt[n]{a^m} = a^{m/n}$ ,  $\sqrt[n]{a} \sqrt[n]{b} = \sqrt[n]{ab}$  or  $a \geq 0$   
**Conjugates:**  $(a + \sqrt{b})(a - \sqrt{b}) = a^2 - b$   
 $\frac{1}{\frac{1}{a} + \frac{1}{b}} = \frac{ab}{a+b}$   
*Always simplify surds and rationalise denominators.*

**QUADRATICS**  
For  $ax^2 + bx + c = 0$ :  
$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$
  
**Discriminant:**  $\Delta = b^2 - 4ac$   
 $\Delta > 0$ : two real roots;  $\Delta = 0$ : repeated root;  $\Delta < 0$ : no real roots.

**COMPLETING THE SQUARE**  
 $ax^2 + bx + c = a\left(x + \frac{b}{2a}\right)^2 + c - \frac{b^2}{4a}$   
For  $x^2 + bx + c$ :  
$$x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + c - \frac{b^2}{4}$$
  
*Use for maximum/minimum values, vertex and inequalities.*

**ROOTS**  
If  $ax^2 + bx + c = 0$  has roots  $\alpha, \beta$ :  
 $\alpha + \beta = -\frac{b}{a}$ ,  $\alpha\beta = \frac{c}{a}$   
Vertex of  $y = ax^2 + bx + c$ :  
$$x = -\frac{b}{2a}$$

**FACTORISATION**  
 $a^2 - b^2 = (a - b)(a + b)$   
 $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$   
 $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$   
 $a^2 + b^2 + c = (x + p)(x + q) + p + q + h$ ,  $px + q$   
*Use for maximum/minimum values, vertex and inequalities.*

**ALGEBRAIC DIVISION AND IDENTITIES**  
If  $f(x) = (x - a)g(x) + r$ , then  $r = f(a)$ .  
An identity is true for all values of  $x$ .  
An equation is true only for its solutions.  
Compare coefficients when two polynomial expressions are identical.

**SIMULTANEOUS EQUATIONS AND INTERSECTIONS**  
Simultaneous method: substitute the linear expression into the quadratic, solve the resulting quadratic, then back-substitute.  
Geometrically: solutions are intersection points of the two graphs.  
A line can meet a quadratic in 0, 1, or 2 points; use the discriminant after substitution.

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2. Inequalities, functions and graphs

**INEQUALITY RULES**  
If  $a < b$ , then  $a + c < b + c$ .  
If  $a > b$ , then  $a + c > b + c$ .  
If  $a < b$ , then  $ac > bc$  Reverse the sign.  
For products/quotients, use critical values and a sign chart.

**QUADRATIC INEQUALITIES**  
To solve  $f(x) > 0$ , find the roots and identify where the graph is above the  $x$ -axis.  
For  $a(x - r_1)(x - r_2)$ , inside the roots if  $a > 0$ , outside the roots if  $a < 0$ .  
Sketch first: it prevents most sign errors.

**MODULUS**  
 $|x| < a \iff -a < x < a$ , for  $a > 0$   
 $|x| > a \iff x < -a$  or  $x > a$ , for  $a > 0$   
 $|x - a|$  means distance from  $a$  on the number line.

**AM-GM**  
For  $a, b \geq 0$ :  
$$\frac{a + b}{2} \geq \sqrt{ab}$$
  
Equality holds if  $a = b$ .  
Useful for maximum-value problems such as  $x + \frac{1}{x}$ .

**FUNCTIONS**  
 $f(g(x))$  means apply  $g$  first, then  $f$ .  
 $\sqrt{x}$  means the positive square root.  
One-to-one: each output comes from at most one input.  
Many-to-one: different inputs can give the same output.

**COMMON GRAPHS**  
• Line:  $y = mx + c$   
• Quadratic:  $y = ax^2 + bx + c$   
• Cubic: possible one, two, or three real roots  
• Reciprocal:  $y = \frac{1}{x}$ , asymptotes  $x = 0$ ,  $y = 0$   
• Exponential:  $y = a^x$ ,  $a > 0$   
• Logarithm:  $y = \log_a x$ ,  $x > 0$   
• Modulus:  $y = |x|$

**GRAPH TRANSFORMATIONS**

| New graph      | Transformation from $y = f(x)$             |
|----------------|--------------------------------------------|
| $y = f(x) + a$ | up $a$                                     |
| $y = f(x) - a$ | down $a$                                   |
| $y = f(x + a)$ | left $a$                                   |
| $y = f(x - a)$ | right $a$                                  |
| $y = af(x)$    | vertical stretch by factor $a$             |
| $y = f(ax)$    | horizontal stretch by factor $\frac{1}{a}$ |
| $y = f(x)$     | reflection in the $x$ -axis                |
| $y = f(-x)$    | reflection in the $y$ -axis                |

**LOGARITHMS**  
 $a^x = c \iff x = \log_a c$   
 $\log_a(xy) = \log_a x + \log_a y$   
 $\log_a\left(\frac{x}{y}\right) = \log_a x - \log_a y$   
 $\log_a(a^x) = x$   
 $\log_a\left(\frac{1}{x}\right) = -\log_a x$   
 $\log_a a = 1$

**EXPONENTIALS**  
 $y = a^x \iff x = \log_a y$   
 $\log_a a^x = x$   
 $a^{\log_a x} = x$   
Change of base:  $\log_a x = \frac{\log_b x}{\log_b a}$   
Change of base is useful but not required by the 2025 spec.

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3. Sequences, series, binomial and counting

**ARITHMETIC SEQUENCES**  
 $u_n = a + (n - 1)d$   
$$S_n = \frac{n}{2}(2a + (n - 1)d) = \frac{n}{2}(a + l)$$
  
First difference is constant.

**GEOMETRIC SEQUENCES**  
 $u_n = ar^{n-1}$   
Finite sum:  
$$S_n = \frac{a(1 - r^n)}{1 - r}, \quad r \neq 1$$
  
Infinite sum:  
$$S_\infty = \frac{a}{1 - r}, \quad |r| < 1$$

**STANDARD SUMS**  
$$\sum_{k=1}^n k = \frac{n(n+1)}{2}$$
  
$$\sum_{k=1}^n c = cn$$
  
*Useful in arithmetic series and proof-style questions.*

**RECURRENCE RELATIONS**  
A recurrence defines terms using previous terms, e.g.  $u_{n+1} = f(n, u_n)$ .  
Generate terms carefully: start from the given  $u_1$  or  $x_0$ .

**LINEAR AND QUADRATIC SEQUENCES**  
Linear sequence: constant first difference.  
Quadratic sequence: constant second difference.  
If  $u_n = an^2 + bn + c$ , then second difference =  $2a$ .

**BINOMIAL EXPANSION**  
$$(a + b)^n = \sum_{r=0}^n \binom{n}{r} a^{n-r} b^r$$
  
$$\binom{n}{r} = \frac{n!}{r!(n-r)!} = \frac{n!}{r!(n-r)!}$$
  
For coefficient of  $x^k$ , find the term where powers combine to  $x^k$ .

**COMBINATIONS**  
 ${}^nC_r = \binom{n}{r} = \frac{n!}{r!(n-r)!}$   
Use when order does not matter.

**PERMUTATIONS**  
 ${}^nP_r = \frac{n!}{(n-r)!}$   
Use when order does matter.

**SYSTEMATIC COUNTING**  
If one task has  $m$  choices and the next has  $n$  choices, the ordered pair of tasks has  $mn$  choices.  
Use lists, grids, tables and tree diagrams to avoid double-counting.  
For repeated choices, check carefully whether repetition is allowed.

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4. Coordinate geometry and circle geometry

**STRAIGHT LINES**  
Gradient:  
$$m = \frac{y_2 - y_1}{x_2 - x_1}$$
  
Point-gradient form:  
 $y - y_1 = m(x - x_1)$   
General form:  $ax + by + c = 0$   
Parallel:  $m_1 = m_2$   
Perpendicular:  $m_1 m_2 = -1$

**DISTANCE AND MIDPOINT**  
$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$
  
$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

**CIRCLES: EQUATIONS**  
 $(x - a)^2 + (y - b)^2 = r^2$   
Centre  $(a, b)$ , radius  $r$ .  
If  $x^2 + y^2 + 2gx + 2fy + c = 0$ :  
centre  $(-g, -f)$ ,  $r = \sqrt{g^2 + f^2 - c}$

**TANGENT TO CIRCLE**  
The tangent at a point is perpendicular to the radius at that point.  
If radius gradient is  $m$ , tangent gradient is  $-1/m$ .  
Find the radius gradient first, then use perpendicular gradient rule.

**CIRCLE THEOREMS**  
• Angle at the centre is twice the angle at the circumference.  
• Angle in a semicircle is  $90^\circ$ .  
• Angles in the same segment are equal.  
• Opposite angles in a cyclic quadrilateral sum to  $180^\circ$ .  
• Tangent is perpendicular to radius at the point of contact.  
• Angle between tangent and chord equals the angle in the alternate segment.  
• Perpendicular from centre to chord bisects the chord.

**ANGLE FACTS**  
Angles on a straight line sum to  $180^\circ$ .  
Angles around a point sum to  $360^\circ$ .  
Vertically opposite angles are equal.  
Triangle angles sum to  $180^\circ$ .  
Exterior angle of triangle = sum of opposite interior angles.

**PARALLEL LINES**  
Corresponding angles are equal.  
Alternate angles are equal.  
Co-interior angles sum to  $180^\circ$ .

**POLYGONS**  
Sum of interior angles of  $n$ -gon:  
$$(n - 2)180^\circ$$
  
Each interior angle of regular  $n$ -gon:  
$$\frac{(n - 2)180^\circ}{n}$$
  
Sum of exterior angles:  $360^\circ$

**PYTHAGORAS**  
 $a^2 + b^2 = c^2$   
In 3D cuboid diagonal:  
$$d = \sqrt{l^2 + w^2 + h^2}$$

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5. Trigonometry, radians and geometry measurement

**RIGHT-TRIANGLE TRIG**  
$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$
  
$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$
  
$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

**CORE IDENTITIES**  
$$\tan \theta = \frac{\sin \theta}{\cos \theta}$$
  
$$\sin^2 \theta + \cos^2 \theta = 1$$
  
$$1 + \tan^2 \theta = \sec^2 \theta$$

**DOUBLE ANGLE**  
$$\sin 2\theta = 2 \sin \theta \cos \theta$$
  
$$\cos 2\theta = \cos^2 \theta - \sin^2 \theta$$
  
$$\cos 2\theta = 1 - 2 \sin^2 \theta = 2 \cos^2 \theta - 1$$

**SINE AND COSINE RULES**  
$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$
  
$$a^2 = b^2 + c^2 - 2bc \cos A$$
  
Area:  
$$\text{Area} = \frac{1}{2} ab \sin C$$
  
Remember the ambiguous SSA case for one rule.

**EXACT VALUES**

| $\theta$      | $0^\circ$ | $30^\circ$           | $45^\circ$           | $60^\circ$           | $90^\circ$ |
|---------------|-----------|----------------------|----------------------|----------------------|------------|
| $\sin \theta$ | 0         | $\frac{1}{2}$        | $\frac{\sqrt{2}}{2}$ | $\frac{\sqrt{3}}{2}$ | 1          |
| $\cos \theta$ | 1         | $\frac{\sqrt{3}}{2}$ | $\frac{\sqrt{2}}{2}$ | $\frac{1}{2}$        | 0          |
| $\tan \theta$ | 0         | $\frac{1}{\sqrt{3}}$ | 1                    | $\sqrt{3}$           | undefined  |

**RADIANS**  
 $180^\circ = \pi$  radians  
$$\theta^\circ = \frac{\pi}{180} \text{ radians}$$
  
$$\theta \text{ radians} = \frac{180^\circ}{\pi}$$

**ARCS AND SECTORS**  
For  $\theta$  in radians:  
arc length =  $r\theta$   
sector area =  $\frac{1}{2} r^2 \theta$   
segment area = sector area - triangle area

**TRIG GRAPHS**  
 $\sin x$ :  $\cos x$ : period  $360^\circ$  or  $2\pi$   
 $\tan x$ : period  $180^\circ$  or  $\pi$   
 $\sin(-x) = -\sin x$ ,  $\cos(-x) = \cos x$

**BEARINGS**  
Bearings are measured clockwise from North.  
Always written with three figures, e.g.  $047^\circ$ ,  $230^\circ$ .  
Draw North lines at every point involved.

**BASIC AREAS**  
Triangle:  $\frac{1}{2}bh$   
Parallelogram:  $bh$   
Trapezium:  $\frac{1}{2}(a + b)h$   
Circle:  $\pi r^2$   
Circumference:  $2\pi r$  or  $\pi d$

**VOLUMES**  
Cuboid:  $lwh$   
Prism: cross-sectional area  $\times$  length  
Cylinder:  $\pi r^2 h$   
Sphere, cone and pyramid formulas are usually given if needed.



6. Differentiation and integration

|                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                               |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>DERIVATIVE MEANING</b></p> $f'(x) = \frac{dy}{dx}$ <p>Derivative = gradient of tangent + instantaneous rate of change.<br/>Second derivative: <math>f''(x) = \frac{d^2y}{dx^2}</math></p>                                                                            | <p><b>POWER RULE</b></p> $\frac{d}{dx}(x^n) = nx^{n-1}$ <p>Works for rational <math>n</math>, where expression is defined.<br/>Simplify first, then differentiate.</p>                                                                                        |
| <p><b>TANGENTS AND NORMALS</b></p> <p>Tangent gradient at <math>x = a</math>: <math>m = f'(a)</math><br/>Tangent: <math>y - y_1 = m(x - x_1)</math></p> <p>Normal gradient: <math>m_{\text{normal}} = -\frac{1}{m}</math></p>                                              | <p><b>STATIONARY POINTS</b></p> <p>Stationary point: <math>f'(x) = 0</math><br/>If <math>f'(x) &gt; 0</math> local minimum.<br/>If <math>f'(x) &lt; 0</math> local maximum.<br/>If <math>f'(x) = 0</math> inspect further.</p>                                |
| <p><b>INCREASING/DECREASING</b></p> <p>If <math>f'(x) &gt; 0</math>, then <math>f</math> is increasing.<br/>If <math>f'(x) &lt; 0</math>, then <math>f</math> is decreasing.<br/><small>Use sign charts for intervals.</small></p>                                         | <p><b>EXTRA DERIVATIVE RULES</b></p> <p><b>xxx</b> Product: <math>(uv)' = u'v + uv'</math><br/><b>xxx</b> Quotient: <math>\left(\frac{u}{v}\right)' = \frac{u'v - uv'}{v^2}</math><br/><b>xxx</b> Chain: <math>\frac{d}{dx}f(g(x)) = f'(g(x))g'(x)</math></p> |
| <p><b>EXTRA STANDARD DERIVATIVES</b></p> <p><b>xxx</b> <math>\frac{d}{dx}e^x = e^x</math><br/><b>xxx</b> <math>\frac{d}{dx}\ln x = \frac{1}{x}</math><br/><b>xxx</b> <math>\frac{d}{dx}\sin x = \cos x</math><br/><b>xxx</b> <math>\frac{d}{dx}\cos x = -\sin x</math></p> | <p><b>POWER INTEGRALS</b></p> $\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$ <p>For definite integrals:<br/><math display="block">\int_a^b f(x) dx = F(b) - F(a)</math></p>                                                                         |
| <p><b>EXTRA STANDARD INTEGRALS</b></p> <p><b>xxx</b> <math>\int \frac{1}{x} dx = \ln x  + C</math><br/><b>xxx</b> <math>\int e^x dx = e^x + C</math><br/><b>xxx</b> <math>\int \cos x dx = \sin x + C</math><br/><b>xxx</b> <math>\int \sin x dx = -\cos x + C</math></p>  | <p><b>FUNDAMENTAL THEOREM</b></p> <p>If <math>F'(x) = f(x)</math>, then:<br/><math display="block">\int_a^b f(x) dx = F(b) - F(a)</math><br/><math display="block">\frac{d}{dx} \left( \int_a^x f(t) dt \right) = f(x)</math></p>                             |
| <p><b>COMBINING INTEGRALS</b></p> $\int_a^b (f(x) + g(x)) dx = \int_a^b f(x) dx + \int_a^b g(x) dx$ $\int_a^b f(x) dx + \int_b^c f(x) dx = \int_a^c f(x) dx$ $\int_a^b f(x) dx = - \int_b^a f(x) dx$                                                                       | <p><b>AREA VS INTEGRAL</b></p> <p>Definite integral can be negative.<br/>Area between curve and x-axis: only positive area.<br/>If curve crosses the axis, split at roots.</p>                                                                                |
| <p><b>TRAPEZIUM RULE</b></p> <p>For equal spacing <math>h</math>:</p> $A \approx \frac{h}{2} [y_0 + y_n + 2(y_1 + \dots + y_{n-1})]$ <p><small>Over/underestimate depends on curvature.</small></p>                                                                        | <p><b>DIFFERENTIAL EQUATIONS</b></p> <p>If <math>\frac{dy}{dx} = f(x)</math>, then:<br/><math display="block">y = \int f(x) dx</math><br/>Use given point/condition to find <math>C</math>.</p>                                                               |



7. Vectors, transformations and similarity

COLUMN VECTORS

$$\begin{pmatrix} a \\ b \end{pmatrix} + \begin{pmatrix} c \\ d \end{pmatrix} = \begin{pmatrix} a+c \\ b+d \end{pmatrix}$$
$$k \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} ka \\ kb \end{pmatrix}$$

VECTOR BETWEEN POINTS

From  $A(x_1, y_1)$  to  $B(x_2, y_2)$

$$\vec{AB} = \begin{pmatrix} x_2 - x_1 \\ y_2 - y_1 \end{pmatrix}$$

Magnitude:

$$\left| \begin{pmatrix} a \\ b \end{pmatrix} \right| = \sqrt{a^2 + b^2}$$

PARALLEL VECTORS

Vectors are parallel if one is a scalar multiple of the other:

$$\mathbf{a} = k\mathbf{b}$$

Collinear points often require parallel vectors.

VECTOR PROOFS

To prove a quadrilateral is a parallelogram, show opposite sides are equal and parallel as vectors.

To prove midpoint, show position vector is average of endpoints.

$$\mathbf{m} = \frac{\mathbf{a} + \mathbf{b}}{2}$$

SIMILARITY

Scale factor  $k$

Lengths scale by  $k$

Areas scale by  $k^2$

Volumes scale by  $k^3$

CONGRUENCE

Triangle congruence tests:

SSS, SAS, ASA, RHS

Congruent shapes have equal corresponding lengths and angles.

TRANSFORMATIONS

Translation: every point moves by the same vector.

Reflection: mirror line is perpendicular bisector between point and image.

Rotation: use centre, angle and direction.

Enlargement: use centre and scale factor.

Negative scale factor: enlargement plus reversal through centre.

QUADRILATERALS

Parallelogram: opposite sides parallel and equal.

Rectangle: parallelogram with right angles.

Rhombus: all sides equal.

Square: rectangle and rhombus.

Kite: two pairs of adjacent equal sides.

Trapezium: one pair of parallel sides.

3D GEOMETRY VOCABULARY

Know: face, surface, edge, vertex, prism, cylinder, cone, sphere, hemisphere, pyramid, plan and elevation.

For 3D Pythagoras, first find a diagonal on one face, then use it in a second right triangle.



8. Number, units, ratio, percentage and bounds

STANDARD FORM

$a \times 10^n, \quad 1 \leq a < 10$

For multiplication, multiply a's and add powers of 10.  
For division, divide a's and subtract powers of 10.

UNITS

Speed:

$$\text{speed} = \frac{\text{distance}}{\text{time}}$$

Density:

$$\rho = \frac{m}{V}$$

Pressure:

$$p = \frac{F}{A}$$

Convert units before calculating.

PRIME FACTORISATION

Write numbers as products of prime powers.  
HCF: take smaller powers common to both.  
LCM: take larger powers appearing in either.

RATIO

For ratio a : b, total parts = a + b.

First share =  $\frac{a}{a+b} \times \text{total}$ .

Second share =  $\frac{b}{a+b} \times \text{total}$ .

Scale factor =  $\frac{\text{part share}}{\text{all parts}}$ .

DIRECT PROPORTION

$y \propto x \iff y = kx$

$y \propto x^n \iff y = kx^n$

Find k using one known pair:

INVERSE PROPORTION

$y \propto \frac{1}{x} \iff y = \frac{k}{x}$

$y \propto \frac{1}{x^n} \iff y = \frac{k}{x^n}$

PERCENTAGES

$\% \text{ change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100$

Increase by  $r\%$ : multiply by  $1 + \frac{r}{100}$ .

Decrease by  $r\%$ : multiply by  $1 - \frac{r}{100}$ .

GROWTH AND DECAY

Compound growth:

$A = P(1 + r)^n$

Compound decay:

$A = P(1 - r)^n$

Here  $r$  is decimal rate, not percentage.

BOUNDS

If  $x$  is rounded to nearest unit:

$x - \frac{1}{2} \leq \text{true value} < x + \frac{1}{2}$

Nearest tenth:  $\pm 0.05$

Nearest hundredth:  $\pm 0.005$

RECURRING DECIMALS

Let  $x$  equal the recurring decimal.

Multiply by 10, 100, or 1000 to align recurring parts.

Subtract equations to remove recurrence.

APPROXIMATION

Use sensible estimates before calculating. In no-calculator TMUA, approximation often eliminates answer choices.

Always track whether the question asks for exact value, approximation, upper bound, lower bound, or error interval.



9. Probability and statistics

|                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                    |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>PROBABILITY SCALE</b></p> <p><math>0 \leq P(A) \leq 1</math></p> <p>Impossible: 0. Certain: 1.</p> <p>Exhaustive outcomes sum to 1.</p>                                                                                                                                                                                                         | <p><b>COMPLEMENT</b></p> <p><math>P(A') = 1 - P(A)</math></p> <p>Use 'not' or 'at least one' problems by complement.</p>                                                                                                                           |
| <p><b>ADDITION RULE</b></p> <p><math>P(A \cup B) = P(A) + P(B) - P(A \cap B)</math></p> <p>If mutually exclusive:</p> <p><math>P(A \cup B) = P(A) + P(B)</math></p>                                                                                                                                                                                   | <p><b>CONDITIONAL PROBABILITY</b></p> <p><math>P(A   B) = \frac{P(A \cap B)}{P(B)}</math>    <math>P(B) &gt; 0</math></p> <p>*Given <i>B</i> means restrict the sample space to <i>B</i></p>                                                       |
| <p><b>INDEPENDENCE</b></p> <p>Independent if:</p> <p><math>P(A \cap B) = P(A)P(B)</math></p> <p>Equivalently:</p> <p><math>P(A   B) = P(A)</math></p>                                                                                                                                                                                                 | <p><b>TREE DIAGRAMS</b></p> <p>Multiply along branches.</p> <p>Add separate routes to the same outcome.</p> <p>Dependent events: second probability changes after first outcome.</p> <p>Independent events: second probability stays the same.</p> |
| <p><b>EXPECTED FREQUENCY</b></p> <p><b>expected frequency</b> = <math>nP(A)</math></p> <p>Relative frequency:</p> <p><math display="block">\frac{\text{number of successful outcomes}}{\text{number of trials}}</math></p>                                                                                                                            | <p><b>VENN COUNTING</b></p> <p><math> A \cup B  =  A  +  B  -  A \cap B </math></p> <p><math> A'  =  U  -  A </math></p> <p>Fill the intersection first.</p>                                                                                       |
| <p><b>AVERAGES</b></p> <p><math display="block">\text{mean} = \frac{\sum x}{n}</math><p><b>range</b> = <math>\text{max} - \text{min}</math></p><p>Median = middle value after ordering.</p><p>Mode = most frequent value.</p></p>                                                                                                                     | <p><b>GROUPED DATA</b></p> <p>Estimated mean:</p> <p><math display="block">\bar{x} = \frac{\sum fx}{\sum f}</math></p> <p>Use class midpoints for <i>x</i>.</p> <p>Modal class = class with highest frequency.</p>                                 |
| <p><b>HISTOGRAMS</b></p> <p><math display="block">\text{frequency density} = \frac{\text{frequency}}{\text{class width}}</math><p><b>frequency</b> = <math>\text{frequency density} \times \text{class width}</math></p></p>                                                                                                                          | <p><b>QUANTILES AND SPREAD</b></p> <p><math>IQR = Q_3 - Q_1</math></p> <p>Use median and IQR to compare skewed data sets.</p>                                                                                                                      |
| <p><b>SCATTER GRAPHS</b></p> <p>Positive correlation: as one variable increases, the other tends to increase.</p> <p>Negative correlation: as one variable increases, the other tends to decrease.</p> <p><b>Correlation does not imply causation.</b></p> <p>Interpolation is within the data range; extrapolation is outside and less reliable.</p> |                                                                                                                                                                                                                                                    |



10. Logic, language and proof

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|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p><b>LOGIC WORDS</b></p> <p>‘<i>A</i> and <i>B</i>’ means both are true.<br/>‘<i>A</i> or <i>B</i>’ means inclusive or: <i>A</i>, <i>B</i>, or both.<br/>‘Not <i>A</i>’ means <i>A</i> is false.<br/>THUA means precise reading of English.</p>                                                                                                                                                                            | <p><b>IF / ONLY IF / IFF</b></p> <p>‘If <i>A</i>, then <i>B</i>’ means <math>A \Rightarrow B</math>.<br/>‘<i>A</i> if <i>B</i>’ means <math>B \Rightarrow A</math>.<br/>‘<i>A</i> only if <i>B</i>’ means <math>A \Rightarrow B</math>.<br/>‘<i>A</i> if and only if <i>B</i>’ means both <math>A \Rightarrow B</math> and <math>B \Rightarrow A</math>.</p> |
| <p><b>CONVERSE AND CONTRAPOSITIVE</b></p> <p>Statement: <math>P \Rightarrow Q</math><br/>Converse: <math>Q \Rightarrow P</math>: not equivalent in general.<br/>Inverse: <math>\neg P \Rightarrow \neg Q</math>: not equivalent in general.<br/>Contrapositive: <math>\neg Q \Rightarrow \neg P</math>: always equivalent.</p>                                                                                              | <p><b>NECESSARY AND SUFFICIENT</b></p> <p>If <math>P \Rightarrow Q</math><br/><i>P</i> is sufficient for <i>Q</i>.<br/><i>Q</i> is necessary for <i>P</i>.<br/>‘You might as well!’: oxygen is necessary for life.</p>                                                                                                                                       |
| <p><b>QUANTIFIERS</b></p> <p>‘For all’ means every case.<br/>‘For some’ means at least one case.<br/>‘There exists’ means at least one example exists.</p>                                                                                                                                                                                                                                                                  | <p><b>NEGATING QUANTIFIERS</b></p> <p><math>\neg(\forall x, P(x)) \Leftrightarrow \exists x, \neg P(x)</math><br/><math>\neg(\exists x, P(x)) \Leftrightarrow \forall x, \neg P(x)</math><br/>‘Not all’ means ‘at least one not’.</p>                                                                                                                        |
| <p><b>DE MORGAN</b></p> <p><math>\neg(P \wedge Q) \Leftrightarrow \neg P \vee \neg Q</math><br/><math>\neg(P \vee Q) \Leftrightarrow \neg P \wedge \neg Q</math><br/>In English: not both + one or both fail.</p>                                                                                                                                                                                                           | <p><b>TRUTH-TABLE FACTS</b></p> <p>Implication: <math>P \Rightarrow Q</math> is false only when <i>P</i> is true and <i>Q</i> is false.<br/>A biconditional <math>P \Leftrightarrow Q</math> is true when <i>P</i> and <i>Q</i> have the same truth value.</p>                                                                                               |
| <p><b>SET NOTATION</b></p> <p>U: union, <math>\cap</math>: intersection, <math>\complement</math>: complement of<br/>C: subset, <math>\emptyset</math>: empty set, <i>A</i>’: complement<br/><math>A \cap (B \cup C) = (A \cap B) \cup (A \cap C)</math></p>                                                                                                                                                                | <p><b>COUNTEREXAMPLES</b></p> <p>To disprove ‘for all <i>x</i>, <i>P</i>(<i>x</i>)’: one counterexample is enough.<br/>To disprove ‘there exists <i>g</i> such that <i>P</i>(<i>g</i>)’: prove no case works.</p>                                                                                                                                            |
| <p><b>DIRECT PROOF</b></p> <p>Start from known assumptions and proceed logically.<br/><math>A \Rightarrow B \Rightarrow C \Rightarrow \dots \Rightarrow Z</math><br/>Each step must follow from earlier steps.</p>                                                                                                                                                                                                          | <p><b>PROOF BY CASES</b></p> <p>Split into exhaustive cases, e.g. even/odd, positive/negative/zero.<br/>Every possible case must be covered.</p>                                                                                                                                                                                                             |
| <p><b>PROOF BY CONTRADICTION</b></p> <p>To prove <i>S</i>: assume <math>\neg S</math>.<br/>Derive an impossibility.<br/>Conclude <i>S</i> must be true.</p>                                                                                                                                                                                                                                                                 | <p><b>PROOF BY CONTRAPOSITIVE</b></p> <p>To prove <math>P \Rightarrow Q</math> prove:<br/><math>\neg Q \Rightarrow \neg P</math><br/>Often easier when <i>Q</i> contains a negative condition.</p>                                                                                                                                                           |
| <p><b>INDUCTION</b></p> <p><b>base</b>: Base case: prove <i>P</i>(1).<br/><b>step</b>: Inductive step: assume <i>P</i>(<i>k</i>), prove <i>P</i>(<i>k</i> + 1).<br/><b>concl</b>: Conclude <i>P</i>(<i>n</i>) for all required integers <i>n</i>.</p>                                                                                                                                                                       | <p><b>COMMON PROOF ERRORS</b></p> <p>Invalid: <math>a \Rightarrow b \Rightarrow c \Rightarrow b \Rightarrow c</math> unless <math>a \neq \emptyset</math>.<br/>Invalid: <math>\text{if } A = \text{then } B \Rightarrow A = B</math> without checking other angle relations.<br/>Invalid: proving examples and claiming the general result.</p>              |
| <p><b>PAPER 2 PANIC CHECKLIST</b></p> <ul style="list-style-type: none"><li>Underline ‘true’, ‘false’, ‘only if’, ‘for all’, ‘some’, ‘not’.</li><li>Try small cases 0, 1, 2: even/odd, positive/negative.</li><li>Look for a counterexample before doing heavy algebra.</li><li>For implications, test the contrapositive and the converse separately.</li><li>No negative marking: eliminate, guess, flag, move.</li></ul> |                                                                                                                                                                                                                                                                                                                                                              |